

Q1: In the Report Section, which AI is used, how does the algorithm verify fake vs genuine reports, which API is used, and why this model?

Kaunsa AI Hai: Rule-Based NLP Agent aur In-Browser Computer Vision (Image Checker).

- **Fake Filter:** Agar description me 'test', 'prank', 'joke', 'haha' likha hai, ya 15 letter se chota message hai, ya photo mobile ka screenshot hai → AI use turant **Fake / Spam** mark kar deta hai.
- **Zameen Physics Check:** Flat zameen (Cuttack) par agar koi 'Landslide' report karega, toh AI use reject kar dega kyunki flat zameen par pahad nahi hote.
- **Points & Thresholds:** Bada message (+25), GPS (+15), photo (+20), mausam match (+20) se score banta hai: **Score 70% se zyada → Genuine (Asli), 40% se 70% → Needs Review, 40% se kam → Fake / Spam.**

Kaunsi APIs: Mobile ka GPS (Geolocation API), Photo checker (Canvas API), aur local storage (IndexedDB).

Why this model: Disaster me internet band ho jata hai. Heavy GPT-4 ya cloud model bina internet fail ho jate. Hamara model browser ke CPU par 1 millisecond me bina internet ke 100% offline chalta hai.

Q2: In the Live Alerts section, how are earthquakes and floods detected, which algorithm is used, and which APIs are integrated?

Kaunsa System Hai: Telemetry Anomaly Detector (Live Sensor Processor).

- **Earthquake (Bhukamp):** USGS API se live Richter magnitude dekhta hai: **Magnitude 5.5 se zyada → Critical Red Alert, 4.0 se 5.4 → Warning Alert.** Sath hi map par impact radius ka circle bana deta hai.
- **Flood (Baadh) & Toofan:** Open-Meteo satellite se live barish aur hawa check karta hai: **Barish 5 mm/hour se zyada → Flash Flood Warning, Hawa 35 km/hour se zyada → Cyclone / Gale Alert.**

Kaunsi APIs: USGS Global Earthquake API aur Open-Meteo Weather API.

Why these APIs: USGS aur Open-Meteo open-science live feeds hain jo 100% free hain, inme koi billing limit nahi hoti aur real sensors ka data milta hai.

Q3: In the Dashboard & Risk Forecast section, how is disaster risk probability calculated, and why use this model instead of simple percentages?

Kaunsa Model Hai: Bayesian Bell Curve (Risk Graph) aur 6-Axis Radar Chart.

- **Bell Curve Graph:** Live mausam ke hisaab se curve ka peak aage-piche shift hota hai: Normal mausam me curve **Low Risk (20-25%)** par rehta hai, lekin barish aur hawa badhne par curve ka peak **High Risk (75-85%)** par shift ho jata hai.
- **Radar Chart:** Ek sath 6 khatron (Flood, Cyclone, Heatwave, Earthquake, Drought, Tsunami) ka live 0 se 100% meter ek hi graph me dikhata hai.

Kaunsi API / Tool: Open-Meteo Weather API aur Recharts library.

Why this model: Sirf '60% barish hogi' bolne ke badle, Bell Curve graph rescue teams ko poora risk spread aur sensor confidence dikhata hai.

Q4: In the Map & Navigation section, how does shortest safe path routing work, which algorithm is used, and what happens when internet goes down?

- **Shortest Safe Path ke liye: Dijkstra / OSRM Algorithm** (ye road network par flooded aur blocked raste ko bypass karke sabse chhota aur safe road rasta nikalta hai).
- **Distance ke liye: Haversine Algorithm** (GPS coordinates se exact kilometer distance calculate karta hai).
- **Offline Mode ke liye:** Agar internet band ho, toh hamara offline curve formula 32 km/h emergency speed ke hisaab se turn-by-turn rasta aur pahuche ka time (ETA) nikal deta hai.

Kaunsi APIs: Project OSRM Routing Machine, Komoot Photon OpenStreetMap API, aur Leaflet Map.

Why Leaflet & OSRM: Google Maps mehenga hai aur offline caching allow nahi karta. Leaflet aur OSRM 100% free hain, open-source hain aur flooded khatron ko bypass karke offline safe path nikalte hain.

Q5: In the Emergency Help section, how are nearby hospitals and shelters found, and how does 1-tap SOS work?

Kaunsa Algorithm Hai: Bounding Box Search + Proximity Sorting (User ke GPS se 15-45km radius me hospitals, police thana, fire brigade, aur shelters dhoondh kar sabse paas wale ko top par sort karta hai).

- **1-Tap SOS:** Direct phone ka dialer open karta hai bina kisi typing ke (Police: 112, Ambulance: 108, Fire: 101) aur sath hi WhatsApp/SMS par exact GPS location share kar deta hai.

Kaunsi APIs: Komoot Photon OSM API, Tel URI Protocol, aur Web Share API.

Why direct protocol: Emergency me victim ke paas type karne ka time nahi hota. 1-Tap direct phone call lagata hai aur location share kar deta hai.

Q6: In the Volunteer Dashboard, how are genuine incidents prioritized and dispatched to volunteers?

Kaunsa Algorithm Hai: Priority Triage Sorting + 6-Step Mission Lifecycle.

- **Triage Rules:** Sabhi fake aur unverified reports ko hata deta hai. Ranking: **Critical (Jaan ka khatra) > Medium > Nearest Proximity.**
- **6-Step Flow:** Report aayi → Claim kiya → EnRoute (Raste me) → OnScene (Pahucha) → Madad di → Resolved (Complete).

Why this model: Volunteers ka time waste nahi hota, fake calls rukti hain aur duplicate rescue teams ek hi jagah nahi jaati.

Q7: In the Offline Data section, how does the app store data during network blackouts and sync when internet comes back?

Kaunsa Architecture Hai: IndexedDB Local Database + Offline Sync Queue.

- **Offline Storage:** Internet na hone par report phone ke IndexedDB me PendingSync tag ke sath save ho jaati hai.
- **Auto Sync:** Jaise hi internet aata hai, OfflineSyncManager background me reports upload karke status Verified kar deta hai.

Kaunsi Browser APIs: IndexedDB v2 (5 Stores) aur Vite PWA Service Worker.

Why IndexedDB why not LocalStorage: LocalStorage sirf 5MB data store kar sakta hai aur phone hang karta hai. IndexedDB 500MB+ data aur photos ko bina kisi lag ke offline store karta hai.

SECTION	KAUNSA MODEL HAI?	KAUNSA ALGORITHM CHAL RAHA HAI?	WHY THIS MODEL?
1. Report Section	Bayesian NLP + Image Checker	Score $\geq$ 70% Genuine, < 40% Spam	Bina internet phone me 1ms me chalta hai.
2. Live Alerts	Earthquake & Flood Detector	USGS Richter $\geq$ 5.5 Critical, Rain $\geq$ 5mm Flood	Real government sensor data, zero false alarm.
3. Risk Forecast	Bayesian Bell Curve + Radar	Live Weather Shift (Low 25% $\rightarrow$ High 85%)	Poor risk spread aur confidence dikhata hai.
4. Disaster Map	OSRM Dijkstra + Sinusoidal	Haversine Distance + Flooded Road Bypass	100% Free aur offline safe rasta deta hai.
5. Volunteer Desk	Priority Triage + 6-Step Flow	Critical > Medium > Nearest Proximity	Fake calls rokta hai, duplicate rescue bachata hai.
6. Offline Sync	IndexedDB + Sync Queue	FIFO Auto Reconnection Sync	500MB+ storage deta hai bina phone hang kiye.